

Word/statement	Definition
Relative atomic mass	The average mass of the naturally occurring atoms of an element on a scale where the carbon-12 atom has a mass of exactly 12 units (relative to an atom of carbon-12)
Isotopes	Atoms of the same element with the same proton number but different mass number
Molecular mass	The sum of the relative atomic masses of the elements in a molecule
Proton number	Number of protons in the nucleus of an atom
Nucleon number	Total number of protons and neutrons in the nucleus of an atom
Radioisotopes	Isotopes with an unstable nucleus
Molecule	Two or more atoms that are sharing electrons and covalently bonded
Compound	Substance made from two or more elements that chemically combined
Element	Substance that is made of atoms of the same atomic number and cannot be split up into anything simpler
Giant covalent structure	A substance where large numbers of atoms are held together by covalent bonds
Giant ionic lattice	A regular pattern made of alternating anions and cations and strong electrostatic forces between
Ion	Particle containing different numbers of protons and electrons as it lost or gained a number of electrons
Molecular formula	The number and type of atoms in one molecule
Relative formula mass	Is used for ionic compound
Empirical formula	Simplest whole number ratio of different atoms or ions in a compound
Ionic bonding	Electrostatic forces between oppositely charged ions
Electrolyte	An aqueous solution or a molten liquid containing ions that decomposes when it conducts electricity and undergoes electrolysis
Electrolysis	The breakdown of an aqueous or molten ionic compound using electric current
Malleable	The ability of elements to be hammered into shape
Alloy	A mixture of a metal with other metals
Excess	More than enough to react as some it reacts. Another reactant is limiting
Catalyst	A substance that speeds up the rate of a chemical reaction by lowering the activation energy without being used up at the end
Equilibrium	The rate of the forward reaction equals the rate of the backwards reaction and the concentration of the reactants and products are constant
Exothermic reaction	Reaction where the energy used to make the bonds is greater than the energy used to break them as making bonds releases energy and breaking them requires energy
Activation energy	The minimum energy that colliding particles must have to react
Redox reaction	Type of reaction where electrons are transferred from one species to another Reaction involving simultaneous oxidation and reduction
Oxidation	Gain of oxygen, increase in oxidation number, and loss of electrons
Oxidizing agent	A substance that is reduced and oxidizes another substance
Reduction	Loss of oxygen, decrease in oxidation number, and gain of electrons
Reducing agent	A substance that is oxidized and reduces another substance
Acid	Proton donor. $\text{pH} < 7$
Base	Proton acceptor. $\text{pH} > 7$
Weak acid	Partially dissociates into ions
Strong acid	Acid that completely exists as ions and dissociates in solution
Precipitate	A solid that forms when two solutions are mixed and react together
Saturated solution	A solution where no more solute can dissolve at a given temperature
Neutral oxide	Will not react with either acids or bases
Amphoteric oxide	Will react with both acids and bases
Hydrated substance	A substance that is chemically combined with water
Anhydrous substance	A substance containing no water
Water of crystallization	The water molecules present in hydrated crystals
Polymer	Large molecule made from many small monomers joined together
Condensation Polymerization	To form a large molecule from two monomers joining with the removal of a small molecule (water)
Substitution reaction	One atom or group is replaced by another atom or group
Addition reaction	Reaction where only one product is formed
Functional group	The atoms or group of atoms that give any molecule its chemical properties
Homologous series	A family of similar compounds
Hydrocarbon	Compound that contains carbon and hydrogen only
Unsaturated hydrocarbon	Not all the bonds are single and contains a double or triple bond
Biodegradable	Can be broken down by microbes and bacteria
Petroleum	A mixture of hydrocarbons

Lead iodide, PbI_2 Silver iodide, AgI Lead oxide, PbO	All are Yellow
Silver bromide, $AgBr$	Cream
Iron (II) salts	Green
Iron (II) Hydroxide, $Fe(OH)_2$	Dirty Green
Iron (III) Salts	Reddish Brown
Iron (III) hydroxide, $Fe(OH)_3$	
Iron (III) chloride	Yellow
Anhydrous copper sulphate, $CuSO_4$	White (produced when $CuSO_4 \cdot 5H_2O$ is heated)
Hydrated copper sulphate $CuSO_4 \cdot 5H_2O$	Blue
Copper Carbonate	Green
Bromine, Br_2	Brown
Nitrogen dioxide, NO_2	
Potassium dichromate (VI)	Orange Green (when reduced by a reducing agent)
Potassium manganate (VII)	Purple Colourless (when reduced by a reducing agent)
Phenolphthalein	Colourless in acid Pink in base.
Methyl Orange	Red in acid Yellow in base
Universal indicator	Red Orange Yellow Green Blue Indigo Violet Acidic ← neutral → Basic
Litmus	Blue in base Red in acid
Aluminum hydroxide, $Al(OH)_3$ Zinc hydroxide, $Zn(OH)_2$ Calcium hydroxide, $Ca(OH)_2$	White ppt. These compounds are formed when NaOH is added to aq. solutions of their salts.
Flourine	Greenish yellow
Chlorine	Yellowish green
Bromine	Brown / orange
Iodine	Aq. form : Red / brown Solid form : Grey / black
Iron (III) oxide, Fe_2O_3	Orange red
Aluminum oxide, Al_2O_3	Grey
Barium sulphate, $BaSO_4$ Calcium oxide, CaO Magnesium oxide, MgO Zinc oxide, ZnO	White Solids
Copper oxide, CuO	Black
Sodium, Potassium, Calcium, Aluminum, Zinc, Silver metals	Silver
Iron	Silver grey
Copper	Pink

How to remember the Reactivity Series?

Please

Stop

Calling

Me

A

Careless

Zebra

Instead

Try

Learning

How

Copper

Saves

Gold

Potassium

Sodium

Calcium

Magnesium

Aluminium

(Carbon)

Zinc

Iron

Tin

Lead

(Hydrogen)

Copper

Silver

Gold

Most reactive



Least reactive